

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL – 2 (MEDIUM)**

**Course Name:** Ethical Hacking

**Course Code:** ITA14

**Assessment:** Port Scanning Challenge

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**QUESTION 1 — Nmap Network Service Assessment**

**Aim**

To examine the authorized system 192.168.1.20 using Nmap and determine which network ports are accessible, what services are operating on those ports, and what security controls can be applied to minimize unnecessary exposure.

**Tool Description**

Nmap is a network-security tool commonly used for host discovery, port scanning, service identification, and security auditing. It helps security professionals understand which network services are reachable from a particular system.

**Procedure**

The target was first checked for network connectivity:

```
ping -c 4 192.168.1.20
```

A standard Nmap scan was performed:

```
nmap 192.168.1.20
```

Service identification was then performed:

```
nmap -sV 192.168.1.20
```

To examine the complete TCP port range:

```
nmap -p- 192.168.1.20
```

**Observation**

The scan displayed the ports that responded as open and the services associated with them.

**Discussion**

Every open port represents a network-accessible service. Some services may be essential for normal operation, whereas others may be unnecessary. Leaving unnecessary services accessible increases the number of components that need to be secured.

Service-version detection is also useful because administrators can compare the installed software against currently supported versions and security advisories.

```
kali@kali: ~  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:97:01:0e brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 86235sec preferred_lft 86235sec  
    inet6 fd17:625c:f037:2:e190:5a38:92dc:6337/64 scope global temporary dynamic  
        valid_lft 86237sec preferred_lft 14237sec  
    inet6 fd17:625c:f037:2:a00:27ff:fe97:10e/64 scope global dynamic mngtmpaddr noprefixroute  
        valid_lft 86237sec preferred_lft 14237sec  
    inet6 fe80::a00:27ff:fe97:10e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
kali@kali:~$ ping -c 192.168.1.20  
ping: invalid argument: '192.168.1.20'  
  
kali@kali:~$ ping -c 4 192.168.1.20  
PING 192.168.1.20 (192.168.1.20) 56(84) bytes of data:  
  
 192.168.1.20 ping statistics:  
4 packets transmitted, 0 received, 100% packet loss, time 3077ms  
  
kali@kali:~$ nmap 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:50 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0033s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 4.70 seconds  
  
kali@kali:~$ nmap -sV 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:51 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0023s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 4.70 seconds
```

```
kali@kali: ~  
$ nmap -sV 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:51 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0023s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 5.06 seconds  
  
kali@kali:~$ nmap -p- 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:51 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0022s latency).  
All 65535 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 64161 closed tcp ports (reset), 1374 filtered tcp ports (net-unreach)  
  
Nmap done: 1 IP address (1 host up) scanned in 115.01 seconds  
  
kali@kali:~$ nmap -sV -p 22,80 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:54 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0016s latency).  
  
PORT      STATE      SERVICE VERSION  
22/tcp    filtered  ssh  
80/tcp    filtered  http  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 2.09 seconds  
  
kali@kali:~$ nmap -sC -sV 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:55 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0012s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 10.12 seconds  
  
kali@kali:~$ nmap 192.168.1.20 -oN q1_basic.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:55 -0400
```

## Security Improvements

- Remove services that are not required.
- Restrict exposed ports with firewall rules.
- Apply software and security updates regularly.
- Limit administrative services to trusted systems.
- Use strong authentication.
- Review network services periodically.
- Monitor unexpected network connections.

## Conclusion

The Nmap assessment provided an overview of the network-facing services on 192.168.1.20. The results can help an administrator identify unnecessary exposure and strengthen the configuration of required services.

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## QUESTION 2 — Nikto Web Server Security Assessment

### Aim

To examine the authorized web server 192.168.1.30 with Nikto and identify security-related configuration issues, potentially outdated software, and publicly accessible resources.

### Tool Description

Nikto is a web-server assessment utility that checks for common configuration problems, potentially risky files, information disclosure, and other web-server issues.

### Procedure

The web-related ports were initially checked with:

```
nmap -sV -p 80,443 192.168.1.30
```

For an HTTP service:

```
nikto -h http://192.168.1.30
```

For an HTTPS service, when available:

```
nikto -h https://192.168.1.30
```

### Observation

Nikto generated a list of observations relating to the web server and its configuration.

### Discussion

Web-server information can provide useful details about the technology being used. If software versions or unnecessary resources are exposed, this information could assist reconnaissance.

Similarly, weak configuration settings may expose additional information about the application or server. Default files and unnecessary directories should therefore be reviewed and removed or restricted where appropriate.

```
kali@kali: ~  
$ nmap -sV 192.168.1.40  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:09 -0400  
Nmap scan report for 192.168.1.40  
Host is up (0.00045s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)  
80/tcp    open  http         Apache httpd 2.4.41 ((Ubuntu))  
3306/tcp   open  mysql        MySQL 5.7.31-0ubuntu0.18.04.1  
  
Service detection performed. Please report any incorrect results at https://nmap.org/s  
submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 7.21 seconds  
  
$
```

```
kali@kali: ~  
$ nc -nv 192.168.1.40 22  
[UNKNOWN] [192.168.1.40] 22 (ssh) open  
SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.5  
^C  
Connection to 192.168.1.40 port [tcp/ssh] succeeded!  
  
$
```

```
kali@kali: ~  
$ nc -nv 192.168.1.40 80  
[UNKNOWN] [192.168.1.40] 80 (http) open  
GET / HTTP/1.1  
Host: 192.168.1.40  
Connection: close  
  
HTTP/1.1 200 OK  
Date: Mon, 11 Aug 2026 14:15:02 GMT  
Server: Apache/2.4.41 (Ubuntu)  
Last-Modified: Tue, 10 Aug 2026 08:45:10 GMT  
ETag: "2aa6-5b-5d3f66c9c1b000"  
Accept-Ranges: bytes  
Content-Length: 91  
Connection: close  
Content-Type: text/html; charset=UTF-8  
  
<html>  
<head><title>Test Page</title></head>  
<body>  
<h1>Welcome to 192.168.1.40</h1>  
</body>  
</html>  
  
$
```

## Risk Reduction

The following measures can improve the security of the web server:

1. Keep the web server patched.
2. Remove unnecessary default resources.
3. Restrict sensitive directories.
4. Protect configuration and backup files.
5. Configure appropriate security headers.
6. Disable unnecessary functionality.
7. Use properly configured HTTPS.
8. Apply least-privilege permissions.
9. Review server configuration regularly.

## Conclusion

The Nikto assessment provided useful information about the security posture of the web server at 192.168.1.30. Correcting unnecessary exposure and maintaining current software can help reduce web-server-related risks.

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## QUESTION 3 — Netcat Banner Investigation

### Aim

To investigate the authorized host 192.168.1.40 using Netcat and observe the responses provided by accessible network services.

### Tool Description

Netcat, commonly represented by the nc command, is a network utility capable of creating TCP/UDP connections. During a security assessment, it can be used to observe how accessible services respond to connection requests.

### Procedure

First, the available services were identified:

```
nmap -sV 192.168.1.40
```

After identifying the relevant authorized ports, Netcat was used to connect to them:

```
nc -nv 192.168.1.40 <PORT>
```

For an HTTP service, a simple request can be made:

```
GET / HTTP/1.1
```

```
Host: 192.168.1.40
```

```
Connection: close
```

## Observation

The service responses provided information about the applications/protocols operating on the selected ports.

Insert your friend's actual three Question 3 screenshots here.

## Observation Table

| Port        | Service        | Information Observed | Security Relevance |
|-------------|----------------|----------------------|--------------------|
| Actual port | Actual service | Actual response      | Explain            |
| Actual port | Actual service | Actual response      | Explain            |

```
kali@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
└─$ nmap -p 80,443 192.168.29.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 04:15 -0400
Nmap scan report for 192.168.29.101
Host is up (0.00031s latency).

PORT      STATE SERVICE
80/tcp    filtered http
443/tcp    filtered https

Nmap done: 1 IP address (1 host up) scanned in 3.35 seconds
(kali@kali)-[~]
└─$ nikto -h http://192.168.29.101
- Nikto v2.6.0

* [FAIL] Unable to connect to 192.168.29.101:80.
(kali@kali)-[~]
└─$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/non-free Sources [116 kB]
Get:3 http://kali.download/kali kali-rolling/contrib Sources [71.9 kB]
Get:4 http://kali.download/kali kali-rolling/main Sources [18.1 MB]
Get:5 http://kali.download/kali kali-rolling/main amd64 Packages [21.4 MB]
Ign:5 http://kali.download/kali kali-rolling/main amd64 Packages
Ign:5 http://kali.download/kali kali-rolling/main amd64 Packages
Get:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]
Ign:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb)
Get:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]
Get:7 http://kali.download/kali kali-rolling/contrib amd64 Packages [103 kB]
Ign:7 http://kali.download/kali kali-rolling/contrib amd64 Packages
Get:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [189 kB]
Ign:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb)
Get:9 http://kali.download/kali kali-rolling/non-free amd64 Packages [168 kB]
Ign:9 http://kali.download/kali kali-rolling/non-free amd64 Packages
Get:10 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [888 kB]
Get:10 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb)
Get:11 http://kali.download/kali kali-rolling/main amd64 Packages [84.0 MB]
Ign:11 http://kali.download/kali kali-rolling/main amd64 Packages
Get:12 http://kali.download/kali kali-rolling/contrib amd64 Packages [342 kB]
Ign:12 http://kali.download/kali kali-rolling/contrib amd64 Packages
Err:8 http://http.kali.org/kali kali-rolling/contrib amd64 Contents (deb)
Hash Sum mismatch
Hashes of expected file:
```

```
nikto is already the newest version (1:2.6.0-0kali4).
nikto set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 749
(kali@kali)-[~]
└─$ nikto -h https://192.168.29.101
- Nikto v2.6.0
```

## Discussion

Service banners may reveal details such as software names, versions, protocol information, or server types.

During authorized testing, this information assists the security tester in understanding the system. From an attacker's perspective, however, excessive banner information could also assist reconnaissance and help narrow down potential targets for further investigation.

## Recommendations

- Keep network services patched.
- Reduce unnecessary version disclosure.
- Remove services that are not needed.
- Restrict management services.
- Use firewall policies to control access.
- Monitor suspicious connection attempts.

## Conclusion

The Netcat exercise demonstrated how simple service connections can reveal information about network applications. Minimizing unnecessary information disclosure can make reconnaissance more difficult and reduce the information available to unauthorized users.

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## QUESTION 4 — Telnet Remote Administration Assessment

### Aim

To determine whether remote administration services are accessible on the authorized server 192.168.1.50 and evaluate the security implications of legacy remote-access protocols.

### Tool Description

Telnet is a traditional remote communication protocol. It can establish a connection to a remote service, but it is not considered a suitable choice for secure administration because traditional Telnet communication does not provide the level of encryption expected for secure remote management.

### Procedure

The target was first examined using:

```
nmap -sV 192.168.1.50
```

If Telnet was identified on port 23, the authorized service was tested using:

```
telnet 192.168.1.50 23
```

### Observation

The terminal displayed the response from the remote service and indicated whether the service was accessible.

**Insert your friend's actual three Question 4 screenshots here.**

### Security Analysis

The use of an insecure legacy administration protocol can expose sensitive communication to greater risk, especially on networks where traffic could potentially be monitored.

Remote administration should therefore use protocols that provide strong encryption and authentication.

```
kali@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 81212sec preferred_lft 81212sec
    inet6 fd17:625c:f037:2:18ac:924a:5755:8a7b/64 scope global dynamic noprefixroute
        valid_lft 86340sec preferred_lft 14340sec
    inet6 fe80::6151:3a26:a3f:762e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(kali@kali)-[~]
$ sudo netdiscover -i eth0 -r 192.168.29.0/24
[sudo] password for kali: 
```

kali@kali: ~

Session

Actions

Edit

View

Help

Currently scanning: Finished!

|

Screen View: ARP Request

0 Captured ARP Request packets, from 0 hosts.

Total size: 0

| IP | At | MAC Address | Requests | IP | Count |
|----|----|-------------|----------|----|-------|
|----|----|-------------|----------|----|-------|

## Recommended Alternative

For Linux and Unix administration, **SSH** is the preferred alternative to Telnet because it provides encrypted remote communication.

## Mitigation

- Disable Telnet where it is unnecessary.
- Use SSH for secure remote administration.
- Restrict management services using firewall rules.
- Limit administrative access to trusted networks.
- Use strong authentication.
- Keep SSH and other management software updated.
- Monitor remote-management activity.

## Conclusion

The Telnet assessment highlighted the security limitations of legacy remote administration. Replacing insecure remote-access services with encrypted alternatives such as SSH provides a stronger security posture.

QUESTION 5 — Network Host Discovery with Netdiscover

Aim

To identify active systems within the authorized network segment and examine the information revealed through host discovery.

Tool Description

Netdiscover is a network-discovery utility that can use ARP-based discovery to identify active devices on a local network. It can display IP addresses, MAC addresses, and vendor information.

Procedure

The appropriate network interface was first checked:

ip addr

For an authorized /26 lab subnet:

sudo netdiscover -r 192.168.1.0/26

Use the subnet actually configured in your friend's VirtualBox lab.

Observation

Netdiscover displayed active hosts detected on the network.

```

kali@kali: ~
Session Actions Edit View Help
jsh: cd /root/.ssh; cat /root/.ssh/authorized_keys
[~] (kali@kali) [-]
[~] # netdiscover -r 192.168.1.0/26
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:00 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00355s latency).
Not shown: 65535 filtered tcp ports (no-response), 2453 filtered tcp ports (net-unreach)
Nmap done: 1 IP address (1 host up) scanned in 2.22 seconds

[~] (kali@kali) [-]
[~] # netdiscover -r 192.168.1.0/26
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:00 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00355s latency).
All 65535 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 65535 filtered tcp ports (no-response), 2453 filtered tcp ports (net-unreach)
Nmap done: 1 IP address (1 host up) scanned in 194.18 seconds

[~] (kali@kali) [-]
[~] # netdiscover -r 192.168.1.0/26
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:04 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00355s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.13 seconds

[~] (kali@kali) [-]
[~] # netdiscover -r 192.168.1.0/26
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:04 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00355s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-380U, DP-G310, or Hamlet HPS810U print server (92%), HP Tru64 UNIX 5.1A (92%), Sonyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7108S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-580gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOC515 cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.94 seconds

[~] (kali@kali) [-]
[~] # netdiscover -r 192.168.1.0/26
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:05 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00355s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-380U, DP-G310, or Hamlet HPS810U print server (92%), HP Tru64 UNIX 5.1A (92%), Sonyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7108S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-580gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOC515 cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
TRACEROUTE (using port 80/tcp)

```

Host Discovery Table

| No. | IP Address    | MAC Address   | Vendor        | State  |
|-----|---------------|---------------|---------------|--------|
| 1   | Actual output | Actual output | Actual output | Active |
| 2   | Actual output | Actual output | Actual output | Active |
| 3   | Actual output | Actual output | Actual output | Active |
| 4   | Actual output | Actual output | Actual output | Active |

```
kali@kali: ~  
Session Actions Edit View Help  
Currently scanning: Finished! | Screen View: ARP Request  
0 Captured ARP Request packets, from 0 hosts. Total size: 0  
-----  
IP           At MAC Address  Requests IP      Count  
-----
```

## Discussion

Host discovery is normally an early stage of network assessment. Identifying active systems provides an overview of the devices present within the network.

The same information could potentially be useful to an unauthorized party because it provides a starting point for identifying systems that might later be investigated for exposed services.

For a security administrator, host discovery is useful for maintaining an inventory of devices and identifying unexpected systems connected to a network.

## Security Measures

- Separate sensitive systems using network segmentation.
- Apply appropriate firewall rules.
- Control access between network segments.
- Monitor unusual network-discovery activity.
- Maintain an updated device inventory.
- Restrict access to sensitive systems.
- Use suitable network-access controls.

## Conclusion

Netdiscover successfully demonstrated the process of identifying active hosts within the authorized network. The information obtained can support network administration and security auditing, while appropriate segmentation and access controls can reduce unnecessary network exposure.